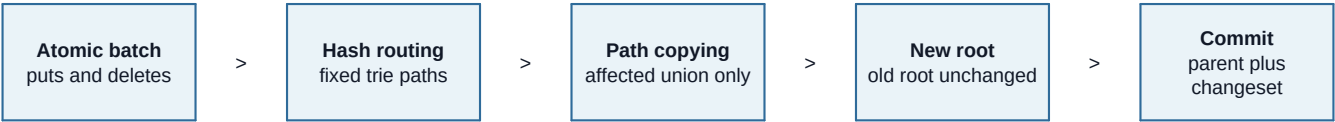


Incremental Merkle Hash Trie

Implemented path copying, addressed history, durable transactions, and adaptive exact differencing.

Commit mechanics



Invariant	Meaning
Incremental update	Every affected leaf is rewritten once; only the union of its ancestor paths is rebuilt.
Structural sharing	Unchanged child hashes are copied into new internal nodes; their subtrees are neither rebuilt nor re-hashed.
Addressed identity	Canonical node, changeset, and commit payloads determine their SHA-256 identities.
Historical stability	Earlier roots and commits remain immutable and continue to reproduce their complete states.
Atomic durability	New objects, version mapping, commit statistics, and HEAD advance in one SQLite transaction.

Geometry and selection

The default geometry is b=8 and d=4. It exposes 4,096 possible leaf buckets and twelve SHA-256 routing bits. Full keys stay in leaves, so keys sharing a routing prefix remain distinguishable.

Policy	Boundary	Purpose
Runtime default	128 operations	Practical default for demos, the API, and repositories without an override.
Audited paper profile	4,096 operations	Frozen from separate calibration before the 333-row evaluation.

What Revon-H reports

Each optimization is paired with an equality check or an explicit work measure.

Correctness contract

- Incremental roots are checked against canonical clean rebuilds in randomized mutation sequences.
- Forced Log, forced Merkle, and Hybrid comparisons must return identical structured differences.
- Every experimental result is checked against a shared deterministic changed-key oracle.
- Repository reopen verifies canonical encoding, object hashes, references, version continuity, metrics, parent links, and HEAD.

Instrumentation

Measure	Interpretation
new trie nodes / bytes	Incremental write amplification for the mutation batch.
reachable-node overlap	Exact diagnostic structural sharing between two roots; excluded from commit timing.
log operations	Natural work unit for changeset aggregation.
node pairs / equal subtrees	Merkle traversal work and pruning effectiveness.
leaf entries	Entry-level scan work below unequal leaf paths.
selected strategy	The path chosen by Revon-H for the requested interval.

Current limits

The implementation has a linear HEAD-only history, one-writer local storage, inline leaf values, and no garbage collection, branches, merges, remotes, schema evolution, or deployment-grade public API security. These are stated limits, not implemented features.

Run the verified flows

```
python -m examples.revon_cli_demo --show-changes
python -m examples.revon_sqlite_demo --database revon_demo.revon.db --reset
python -m unittest -v
```